

1.2. ARITHMETIC RETURNS

We can express a gross return as an *arithmetic return*, which I denote through the book as a lower case r_{t+1} :

$$\text{Arithmetic return } r_{t+1} = R_{t+1} - 1. \quad (\text{A.2})$$

This is the most popular way of expressing rates of return. Arithmetic returns are also called “*simple returns*” or “*net returns*.” Often we just drop any modifier and say just “returns” (as I do throughout the book).

Arithmetic returns are *not* additive over time. Assume the gross return $R_{t+1} = 1.10$. If an asset earns arithmetic returns of $r_{t+1} = (1.10) - 1 = 10\%$ for three periods, then we earn

$$\begin{aligned} W_{t+3} &= (1 + r_{t+1})(1 + r_{t+2})(1 + r_{t+3}) \\ &= 1.10 \times 1.10 \times 1.10 = 1.3310 \\ &\neq 1 + r_{t+1} + r_{t+2} + r_{t+3} \end{aligned}$$

at the end of three periods. We cannot add arithmetic returns. That is $1.3310 \neq 1 + 3 \times 0.10$, because of the compounding of wealth. Even though arithmetic returns are not additive, in some of the figures in this book, I simply cumulate arithmetic returns—to remove the effects of compounding for aesthetics.

Arithmetic returns are, however, additive across assets. Suppose that asset A has a return of 10%, $r_{t+1}^A = 0.10$ and asset B has a return of 5%, $r_{t+1}^B = 0.05$. Then, if we hold a portfolio of 50% asset A and 50% asset B, the arithmetic portfolio return is

$$r_{t+1}^p = 0.5r_{t+1}^A + 0.5r_{t+1}^B = 0.5 \times 0.10 + 0.5 \times 0.05 = 7.5\%.$$

so arithmetic returns conveniently aggregate in a portfolio.

1.3. LOG RETURNS

Log returns, or *continuously compounded returns*, are defined as the natural logarithm of gross returns and I denote them with an upper bar:

$$\text{Log return } \bar{r}_{t+1} = \ln(R_{t+1}) = \ln(1 + r_{t+1}). \quad (\text{A.3})$$

Or we can equivalently state everything in terms of gross returns:

$$R_{t+1} = \exp(\bar{r}_{t+1}) = 1 + r_{t+1}.$$